

# Misr University for Science & Technology

Faculty of Engineering

Computer & Software Engineering Department

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## CSE 301 — Database Systems

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### Semester Project Guidelines

Academic Year Spring – 2026

#### Under Supervision Of

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**Total Marks: 15 | Bonus: +5 (Top 3 Projects)**

# 1. Introduction

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This document outlines the guidelines, requirements, evaluation criteria, and deliverables for the CSE 301 Database Systems semester project at Misr University for Science & Technology (MUST). The project is designed to provide students with comprehensive, hands-on experience in designing, implementing, and deploying a complete database-driven application from scratch.

Students will progress through the entire Database Life Cycle (DBLC), starting from system analysis and requirements gathering, through conceptual and logical design, to physical implementation and application integration. This project simulates a real-world software development workflow and emphasizes both teamwork and individual accountability.

## 2. Project Objectives

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- Apply the complete Database Life Cycle (DBLC) methodology on a real-world scenario.
- Conduct proper system analysis and produce professional requirement documentation.
- Design an Entity-Relationship (ER) Diagram that accurately captures the system domain.
- Perform ER-to-Relational mapping following standard mapping algorithms.
- Implement a physical database on Microsoft SQL Server using SQL DDL and DML commands.
- Develop a functional frontend and backend application integrated with the database.
- Strengthen teamwork, communication, and individual problem-solving skills.

## 3. Team Formation & Composition

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Each project team shall consist of exactly 8 (eight) students. Teams are expected to self-organize, but the following guidelines must be observed:

1. Each team must submit a Team Formation Sheet (template will be provided) containing full names, student IDs, contact information, and assigned roles for each member.
2. Suggested roles include (but are not limited to): System Analyst, Database Designer, Backend Developer, Frontend Developer, Documentation Lead, and Integration Engineer.
3. Team formation sheets must be submitted by the deadline of 10 April 2026.
4. Role reassignment during the project is allowed but must be documented and communicated to the supervisors.

**⚠ Important:** Every team member **MUST** participate in **ALL** project phases. Evaluation is individual, not collective. A member who only contributes to one phase will be graded solely on their demonstrated knowledge across all phases during the oral defense.

## 4. Project Phases (DBLC)

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### Phase 1: System Analysis & Requirement Documentation

Each team must select a real-world domain (e.g., Hospital Management, Library System, E-Commerce Platform, University Registration, Hotel Booking, etc.) and perform thorough system analysis.

### Phase 2: ER Diagram Design

Based on the requirements document, design a comprehensive conceptual Entity-Relationship (ER) Diagram. The ER diagram must include:

- All entities with their attributes (including primary keys, partial keys, derived and multivalued attributes where applicable)
- Relationships between entities with correct cardinality ratios (1:1, 1:N, M:N)
- Participation constraints (total vs. partial)
- Weak entities and identifying relationships (if any)
- Specialization/Generalization hierarchies (if applicable)
- The diagram must be created using a professional tool (e.g., draw.io, Lucidchart, ERDPlus, Microsoft Visio)

### Phase 3: ER-to-Relational Mapping

Apply the standard mapping algorithm to convert the ER diagram into a relational schema. The mapping document must clearly show:

1. Mapping of regular (strong) entity types
2. Mapping of weak entity types
3. Mapping of binary 1:1 relationship type
4. Mapping of binary 1: N relationship types
5. Mapping of binary M: N relationship types
6. Mapping of multivalued attributes
7. Mapping of N-Ary relationship types (if applicable)
8. Mapping of specialization/generalization (if applicable)

The final relational schema must list all tables with their columns, primary keys, foreign keys, and any additional constraints (NOT NULL, UNIQUE, CHECK, DEFAULT).

### Phase 4: Physical Database Implementation

Create the physical database on Microsoft SQL Server. This phase involves:

- Creating the database instance on SQL Server (local or cloud-based)
- Writing SQL DDL scripts (CREATE DATABASE, CREATE TABLE, ALTER TABLE) to implement the full relational schema
- Defining all constraints: primary keys, foreign keys, NOT NULL, UNIQUE, CHECK, and DEFAULT values
- All SQL scripts must be well-commented and submitted as .sql files

## Phase 5: Sample Data & SQL Queries

Populate the database with meaningful sample data and demonstrate SQL proficiency:

1. Insert at least 15-20 records per main entity table using INSERT statements
2. Write at least 10 meaningful SELECT queries demonstrating: joins (INNER, LEFT, RIGHT), subqueries, aggregate functions (COUNT, SUM, AVG, MAX, MIN), GROUP BY and HAVING clauses, and ORDER BY
3. Include at least 2 UPDATE and 2 DELETE operations with WHERE clauses
4. Create at least 2 Views and 2 Stored Procedures
5. All queries must be documented with comments explaining their purpose

## Phase 6: Frontend & Backend Development

Develop a complete application with a user interface (frontend) and server-side logic (backend). There is no mandatory framework or programming language; teams are free to choose their technology stack.

Examples include but are not limited to:

| Frontend              | Backend               |
|-----------------------|-----------------------|
| HTML/CSS/JS           | Node.js (Express)     |
| React / Angular / Vue | Python (Flask/Django) |
| Windows Forms / WPF   | C# (ASP.NET)          |
| Java Swing / JavaFX   | Java (Spring Boot)    |
| PHP (Blade/Twig)      | PHP (Laravel)         |

The application must support at minimum:

- CRUD operations (Create, Read, Update, Delete) for at least 3 core entities
- A login/authentication system (basic level is acceptable)
- Data validation on both client and server side
- Clean, user-friendly interface design
- Proper error handling and meaningful feedback messages

## Phase 7: Integration & Deployment

The final phase involves connecting all three layers (Frontend, Backend, Database) into a cohesive, working system:

- The frontend must communicate with the backend via API calls or direct method invocations
- The backend must perform all database operations through proper connection strings to SQL Server
- Demonstrate a complete user workflow: e.g., user registers, logs in, performs operations, and data persists in the database
- The integrated system must be demonstrated live during the oral defense session

## 5. Evaluation & Grading Criteria

The project carries a total of 15 marks distributed as follows:

| Evaluation Criteria                            | Marks     |
|------------------------------------------------|-----------|
| Requirement Documentation (System Analysis)    | 1         |
| ER Diagram Design                              | 1.5       |
| Mapping ER to Relational Schema                | 1.5       |
| Physical Database Implementation (SQL Server)  | 2         |
| SQL Schema Creation & Sample Data              | 2         |
| Frontend Development                           | 2         |
| Backend Development                            | 2         |
| Database Integration (Frontend + Backend + DB) | 1.5       |
| Individual Contribution & Oral Defense         | 1.5       |
| <b>Total</b>                                   | <b>15</b> |
| <b>Bonus (Top 3 Projects)</b>                  | <b>+5</b> |

### Individual Evaluation Policy

Evaluation is conducted individually within each group. During the oral defense, each student will be asked to:

1. Explain any phase of the project (not just their assigned role)
2. Answer technical questions about the ER diagram, mapping decisions, SQL queries, and code logic
3. Demonstrate their direct contributions to the project
4. Show evidence of their involvement

 **Bonus:** The top 3 projects (based on overall quality, complexity, creativity, and presentation) will receive an additional 5 bonus marks added to the total project grade.

## 6. Required Deliverables

Each team must submit the following deliverables by the final deadline:

| # | Deliverable                            | Description                                                                                |
|---|----------------------------------------|--------------------------------------------------------------------------------------------|
| 1 | <b>Team Formation Sheet</b>            | Names, IDs, roles, and contact info of all 8 members                                       |
| 2 | <b>Requirement Documentation (SRS)</b> | Full system analysis: scope, functional & non-functional requirements, use cases           |
| 3 | <b>ER Diagram</b>                      | Conceptual design with entities, attributes, relationships, cardinalities, and constraints |
| 4 | <b>Relational Schema Document</b>      | Mapped relational tables with primary keys, foreign keys, and normalization notes          |

|   |                                 |                                                                          |
|---|---------------------------------|--------------------------------------------------------------------------|
| 5 | <b>SQL Scripts</b>              | DDL (CREATE statements), DML (INSERT sample data), and sample queries    |
| 6 | <b>Source Code</b>              | Complete frontend and backend source code with clear folder structure    |
| 7 | <b>Final Project Report</b>     | Comprehensive document covering all phases with diagrams and screenshots |
| 8 | <b>Live Demo / Presentation</b> | Working demo of the integrated system during oral defense                |

## 7. Submission Guidelines

1. All deliveries must be submitted electronically via university mail to all Supervision Team.
2. The final report must be in PDF or DOCX format with proper formatting, diagrams, and screenshots.
3. Late submissions will incur a penalty of 1 mark per day (up to 3 days), after which the submission will not be accepted.
4. Plagiarism of any kind (copied code, copied documentation, identical ER diagrams between groups) will result in a zero grade for all parties involved.

## 8. Important Notes & Academic Integrity

- Each student is expected to understand and be able to explain every part of the project, regardless of their assigned role.
- Using AI tools (e.g., ChatGPT, GitHub Copilot) is permitted for learning and assistance, but students must fully understand and be able to explain any AI-generated code or content during the oral defense.
- Oral defense is mandatory. Absence without a valid, documented excuse will result in a zero grade.

***We wish all teams the best of luck!***

### **Supervision Team**

Prof. Dr. Abdulhameed Sharaf | Eng. Bilal Mohamed Elhlwany | Eng. Joumana Emad